

Latent Volatility Estimation through Bayesian Methods

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LSE — ST451 Bayesian Machine Learning

Contents

1	Introduction	1
1.1	Volatility	1
2	Methodology and Models	1
2.1	Rolling Volatility	2
2.2	Bayesian EGARCH(1,1) with t distribution	2
2.3	Standard Stochastic Volatility Model (SV)	2
3	Priors	3
4	Metrics	4
4.1	QLIKE Loss	4
4.2	Realized Volatility Coverage	5
4.3	Return Calibration Test	5
5	Regression	6
5.1	Results	6
5.1.1	QLIKE Loss	6
5.1.2	Realized Volatility Coverage	6
5.1.3	Return Calibration Test	7
6	Sequential Updating	7
6.1	Results	7
6.1.1	QLIKE Loss	8
6.1.2	Realized Volatility Coverage	8
6.1.3	Return Calibration Test	9
7	Conclusions	9
7.1	Project Limitations	9
7.2	Conclusions	10

1 Introduction

This project aims to estimate and identify the latent volatility trends in the market. In particular, credible intervals for latent volatility are generated through Bayesian inference methods using daily S&P 500 market data.

Accurately estimating the latent volatility is particularly useful for market traders. In particular, traders can dynamically hedge their trading positions where theoretically the size of their position should be inversely proportion to volatility. Additionally, they can potentially switch strategies depending on if the latent volatility passes a certain threshold (or a certain credible interval). Furthermore, they can easily identify mispriced options since quoted options generally have an associated implied volatility. All of these techniques can be used to generate a profit.

1.1 Volatility

Volatility is defined as the standard deviation of the daily log returns.

Note that, there is no stated volatility published by the stock market at the end of the day unlike the stock price making it an unobservable 'latent' quantity.

Nonetheless, the effectiveness of a latent volatility model can be tested using the monthly realized volatility (Andersen et al., 2003) (standard deviation over the next 21 log returns).

In particular,

$$\hat{\sigma}_t^{empirical} = \sqrt{\frac{1}{h-1} \sum_{i=1}^h (r_{t+i} - \bar{r}_t)^2}$$

where $r_t = \ln(\frac{P_t}{P_{t-1}})$ and $\bar{r}_t = \frac{1}{h} \sum_{i=1}^h r_{t+i}$ and $h = 21$

is compared with

$$\hat{\sigma}_t^{model} = \sqrt{\frac{1}{h-1} \sum_{i=1}^h \sigma_{t+i}^2}$$

where σ_t is the model's instantaneous volatility estimate at time t .

2 Methodology and Models

Throughout this project daily S&P 500 financial data from Yahoo Finance is used. The 2013-2017, 2018-2021, 2022-2025 time horizons are used for developing priors, testing regression and testing sequential updating respectively.

In this project, 3 different types of models are used (a baseline model, a time series model and a stochastic model) .

2.1 Rolling Volatility

The rolling volatility is the sample standard deviation of the previous h log returns:

$$\hat{\sigma}_t^{roll} = \sqrt{\frac{1}{h-1} \sum_{i=0}^{h-1} (r_{t-i} - \bar{r}_t)^2}$$

where $\bar{r}_t = \frac{1}{h} \sum_{i=0}^{h-1} r_{t-i}$ is the rolling mean and $h = 21$

This method serves as a naive baseline since it is independent of any model assumptions and is quickly computable from past returns.

2.2 Bayesian EGARCH(1,1) with t distribution

The EGARCH(1,1)-t model (Nelson, 1991) is parameterized by:

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad z_t \stackrel{\text{i.i.d.}}{\sim} t_\nu(0, 1) \quad (1)$$

$$\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2 \quad (2)$$

Its key differentiator over other time series models is its ability to capture the leverage effect (the principle that volatility reacts asymmetrically to equal changes in positive and negative returns) through γ .

It also captures other key properties of volatility (the t-distribution captures fat-tailedness of returns and α captures the effect of sudden large changes in price).

The Bayesian nature of the model is visible in its implementation. Essentially, priors for the model parameters are determined from prior data and then the posterior samples are obtained using Markov Chain Monte Carlo Sampling (MCMC) through NumPyro(Phan et al., 2019) (a fast algorithm that optimizes parameter proposals using gradients).

2.3 Standard Stochastic Volatility Model (SV)

The Standard Stochastic Volatility Model (SV) (Taylor, 1986) is a simplistic stochastic model parameterized by:

$$r_t = \mu_r + \sigma_t \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1) \quad (3)$$

$$\ln \sigma_t^2 = \mu_h + \phi(\ln \sigma_{t-1}^2 - \mu_h) + \eta_t, \quad \eta_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2) \quad (4)$$

Unlike standard time series models, where σ_t is only calculated using information at time $t - 1$, this model allows randomness at time t through η_t (independent stochastic volatility shocks), which more directly factors into the uncertainty with latent volatility.

On the other hand, this model doesn't factor in many of the volatility properties that EGARCH does.

This method is implemented using a particle MCMC (Andrieu et al., 2010) since the standard MCMC method isn't computationally feasible here. This is because at time t , all the previous η_t values need to be sampled as they all directly affect the conditional posterior distribution of σ_t .

The particle MCMC simply applies the MCMC method to the particle filter to produce the required posterior samples. This particle filter consists of multiple particles representing $\ln \sigma_t^2$ that are re-weighted and re-sampled at each time step based on the likelihood that the particle's implied volatility can cover the actual returns.

3 Priors

Theoretically priors shouldn't matter much as the likelihood should dominate the prior, but in practice these complex models require good priors for better convergence.

Additionally, these priors need to take the model's stationarity conditions into account; otherwise the models aren't theoretically valid for predictions. (Stationarity suggests that the model is stable i.e that a long-run σ_∞^2 exists).

For Bayesian EGARCH, the priors used are listed below:

$$\begin{aligned} \mu &\sim \mathcal{N}(\hat{\mu}, 0.001^2) & \alpha &\sim \mathcal{N}_{[0, \infty]}(\hat{\alpha}, 0.05^2) & \gamma &\sim \mathcal{N}(\hat{\gamma}, 0.05^2) \\ \beta &\sim \text{Beta}(47, 3) & \nu - 2.1 &\sim \text{Gamma}\left(\frac{\nu - 2.1}{2}, 0.5\right) & \sigma_\infty &\sim \mathcal{N}_{[0, 0.02]}(\sigma_{\text{prior}}, 0.003^2) \end{aligned}$$

where $\omega = (1 - \beta) \log \sigma_\infty^2$ is derived by applying expectations to equation (2) as $t \rightarrow \infty$.

For the stochastic volatility model, the priors used are listed below

$$\mu_h \sim \mathcal{N}(\hat{\mu}_h, 1^2) \quad \phi \sim \text{Beta}(7, 1) \quad \sigma_\eta \sim \mathcal{N}^+(0, 0.5^2) \quad \mu_r \sim \mathcal{N}(\hat{\mu}_r, 0.001^2)$$

where $\mu_h = \log \sigma_\infty^2$ is derived by applying expectations to equation (4) as $t \rightarrow \infty$.

The general idea for developing priors is:

- Use volatility knowledge and stationarity conditions to decide on distribution types
- Use MLE fits (or approximations for the SV model) to determine mean parameters
- Test variances through prior predictive testing

An example of the output from prior predictive testing is given below.

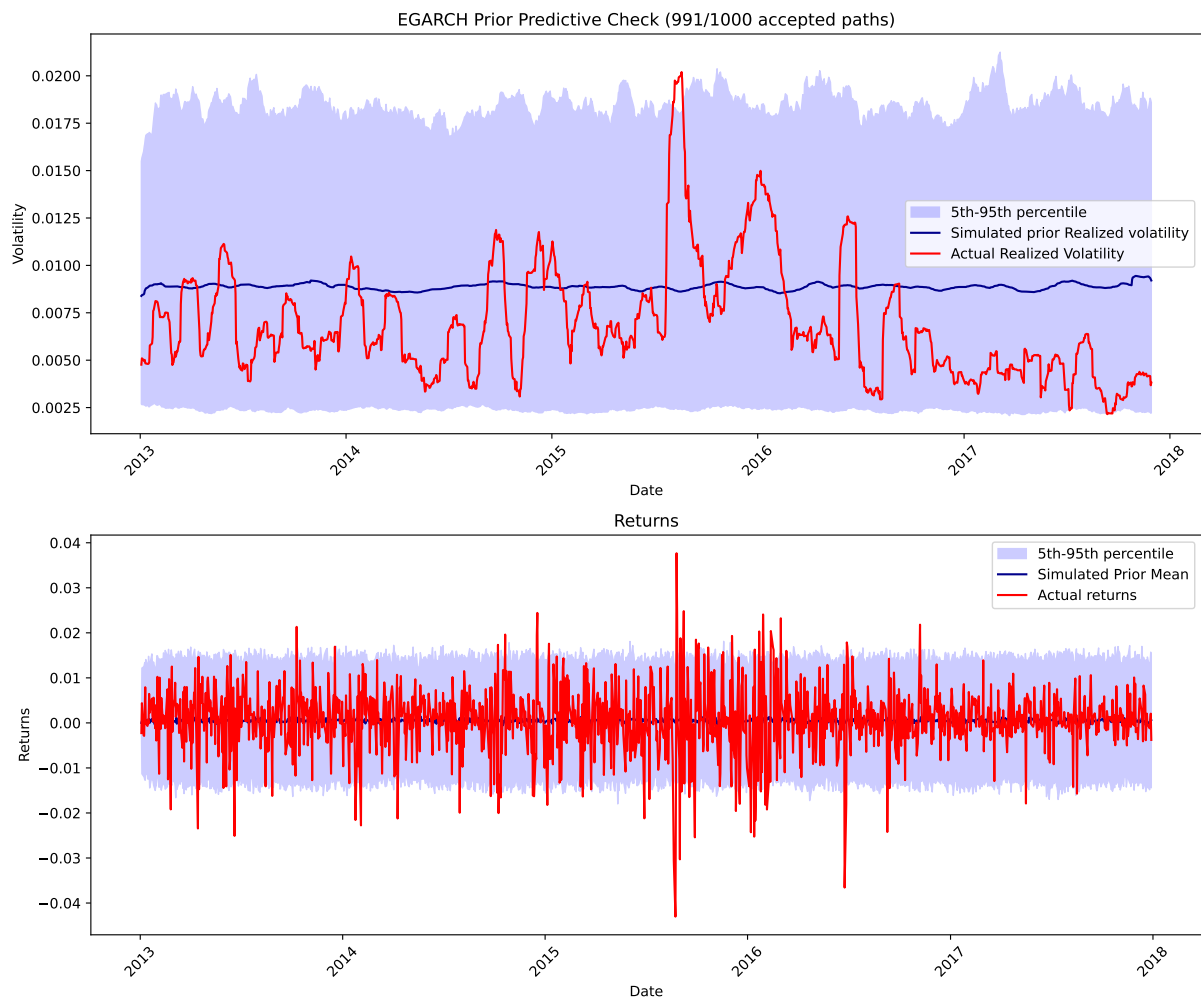


Figure 1: Prior Predictive Testing

The goal is for the percentile bands to cover most of the realized volatility and log returns area without being implausibly wide.

4 Metrics

4.1 QLIKE Loss

This is a straightforward test that can evaluate the baseline model without confidence intervals

Here, the volatility forecasts are evaluated against realized volatility using the QLIKE loss function (Patton, 2011).

$$\text{QLIKE}(\hat{\sigma}_t, RV_t) = \frac{RV_t^2}{\hat{\sigma}_t^2} - \log\left(\frac{RV_t^2}{\hat{\sigma}_t^2}\right) - 1$$

Simply put, a lower mean QLIKE loss is better since it suggests the model closely fits the realized volatility.

4.2 Realized Volatility Coverage

This method directly tests if the model's volatility estimates are suitable.

As discussed in the volatility section $\hat{\sigma}_t^{\text{empirical}}$ which is derived through estimates of the actual realized volatility is compared to $\hat{\sigma}_t^{\text{model}}$ which measures actual realized volatility.

For a legitimate comparison with the empirical, the standard deviation error (noise estimate) needs to be factored into the true model. For simplicity, a log normal model is used. (Patton, 2011)

$$\hat{\sigma}_t^{\text{empirical}} = \sigma_t^{\text{true}} \cdot \exp(\eta_t), \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$$

Hence, the noise estimate computed from the realized volatilities is:

$$\hat{\sigma}_\eta = \text{std} \left(\log \hat{\sigma}_t^{\text{empirical}} - \log \hat{\sigma}_t^{\text{model}} \right) \quad (5)$$

This may seem circular, but it is a sensible approach for a regression task that won't be circular in the sequential section.

Then, the realized volatility coverage is simply the percentage of actual realized volatility estimates that fall within the model's Bayesian credible intervals.

4.3 Return Calibration Test

Here, the actual returns are compared to the estimated distribution of returns derived from the model's volatility estimates through the Probability Integral Transform (PIT) framework (Diebold et al., 1997).

This framework suggest that the fraction of the model's estimated returns that lie below the actual returns at each time t should be uniformly distributed over the entire time period.

This uniformity is theoretically tested using the Kolmogorov-Smirnov (KS) test (Massey, 1951) where $p < 0.05$ suggests that the model is not well calibrated.

Furthermore, a return coverage (in a similar style to the realized volatility coverage) can be extracted for different Bayesian credible intervals as well.

This method is independent of the noise correction in realized volatility but doesn't directly check the volatility estimates making it a supporting metric.

5 Regression

Here, the priors based on 2013-2017 are updated using data from 2018-2021 to obtain posterior samples for the model parameters. These parameters are then propagated forward through the regression data to obtain implied volatility paths, which can be utilized for the realized volatility comparison.

This is an in-sample test that checks whether the models are suited to volatility sampling.

5.1 Results

Throughout this section, the COVID spike is treated as a 3 month period starting from 30th January 2020 (World Health Organization, 2020). This COVID spike gives some insight into how the models react to sudden regime changes (non-stationarity conditions).

5.1.1 QLIKE Loss

Table 1: Mean QLIKE Loss Comparison

Model	Overall	Pre-COVID	COVID Spike	Post-COVID
Rolling	1.2913	1.2380	6.3625	0.5659
EGARCH	0.1073	0.1231	0.1383	0.0819
SV	0.0676	0.0678	0.1794	0.0498

From the QLIKE tests, it is clear that both models are significantly better than the baseline throughout.

Interestingly, SV has a lower overall mean QLIKE loss than EGARCH. In contrast it performs worse than the EGARCH model during the COVID period, which suggests that the SV model is not suited to estimating sudden spikes in volatility.

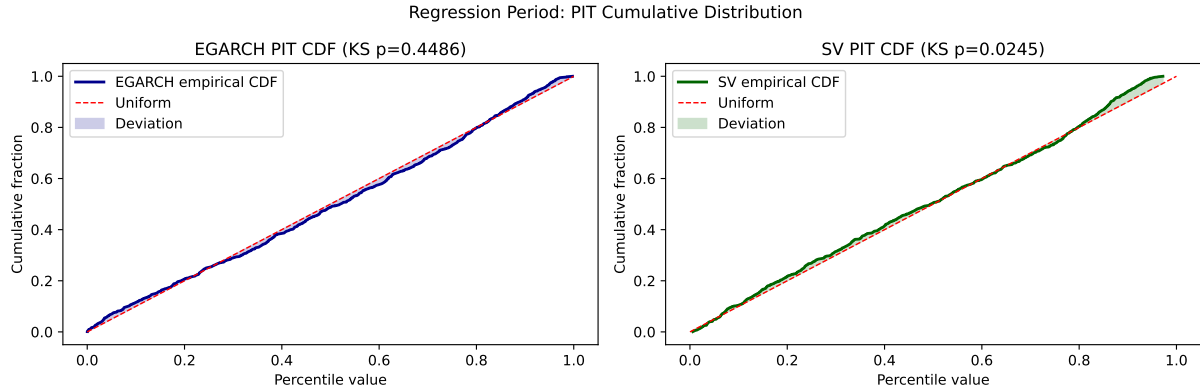
5.1.2 Realized Volatility Coverage

Table 2: Realized Volatility Coverage

Model	80%	85%	90%	95%	99%	Noise $\hat{\sigma}_\eta$
EGARCH	70.21%	77.30%	84.30%	90.17%	96.96%	0.2052
SV	82.47%	86.63%	89.67%	93.92%	98.28%	0.1783

Taking the 4 year period as a whole, the SV model is consistently better at covering realized volatility despite struggling during COVID, since its more representative of the true credible intervals across several thresholds with a lower noise estimate.

5.1.3 Return Calibration Test



From the p values, KS clearly rejects the SV model suggesting that it doesn't accurately reflect the return distribution's shape.

The graph pictorially represents why this happens. Essentially, the SV model's credible intervals cover more returns than they should for higher percentiles suggesting its credible return intervals are too wide.

6 Sequential Updating

Starting with the posterior samples from the regression, the models are sequentially updated over 6 months windows from 2022-2025.

In each window, the volatility is forecasted using the previous window's posterior samples. Then the posterior is updated by treating the previous windows posterior as the prior (this is done by fitting the posterior samples to the distributions of the original priors) and conditioning on the window's data.

Furthermore, the measurement noise is now estimated using all of the realized volatility data prior to the window (including the regression data) through equation (5).

Therefore, this is a truly out-of-sample test since both the model's realized volatility and measurement noise for a window are only based on data prior to the window.

6.1 Results

Note that a discrepancy in the results of the regression section and this section is a sign of model overfitting.

6.1.1 QLIKE Loss

Table 3: Mean QLIKE Loss by Window

Period	Rolling	EGARCH	SV
2022 H1	0.2578	0.0155	0.0170
2022 H2	0.1093	0.0487	0.0429
2023 H1	0.0842	0.1597	0.2396
2023 H2	0.1296	0.1715	0.1690
2024 H1	0.1702	0.0995	0.1231
2024 H2	0.9491	0.0461	0.1334
2025 H1	1.0283	0.1379	0.4826
2025 H2	0.4464	0.1368	0.0741
Total	0.3955	0.1010	0.1607

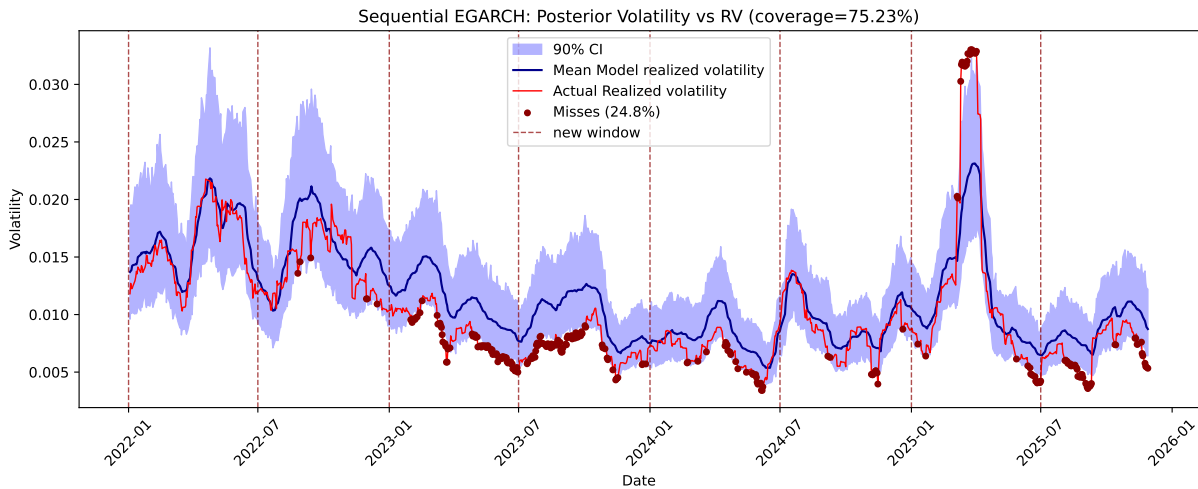
Only the mean QLIKE loss for EGARCH is consistent with regression observations. The mean loss value decreased by more than half for rolling volatility and more than doubled for the SV model.

As expected, the rolling baseline is still significantly worse than the other models although the year 2023 is a surprising exception.

Furthermore, the stochastic volatility model particularly struggles to form accurate predictions in the first half of 2025.

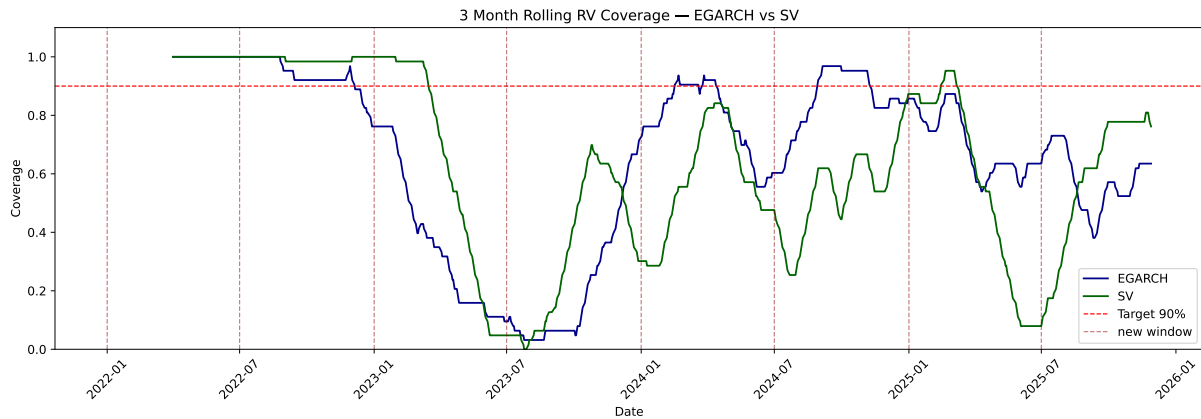
6.1.2 Realized Volatility Coverage

The realized volatility coverage for both the EGARCH and SV models decreased in this out-of-sample test. In particular, they dropped from 84.30% and 89.67% to 75.23% and 74.11% respectively for the 90% credible interval.



The graph above gives further intuition into the QLIKE results and lower realized volatility

coverage. Essentially, the realized volatility was surprisingly low during 2023 and spiked during the first half of 2025, resulting in the red dots visible in the diagram.



The rolling coverage graph highlights the fact that both EGARCH and SV couldn't cover the low realized volatility in the first half of 2023. However, the SV model's realized volatility credible intervals were also unable to cover the sudden spike in 2025 re-emphasizing the SV model's inability to react to more volatile periods.

Note that in most normal periods (like 2024), the coverage of both graphs rise around the start of a new window which makes sense since the model is refit at that point. This suggests that the coverage could improve with smaller windows i.e more refits.

6.1.3 Return Calibration Test

In contrast with realized volatility, there aren't any significant discrepancies with regression in this section. The models have good coverage (96.01 and 95.81% respectively for the 95% returns credible interval). Additionally, the KS results hold (EGARCH is accepted with a 0.1411 p-value and SV is rejected with a 0.0363 p-value).

7 Conclusions

7.1 Project Limitations

In practice, real financial practitioners purchase 5 minute price ticks (Andersen et al., 2003) in order to get better realized volatility estimates and can consequently update their parameters significantly more frequently.

This access to more data allows models to be refit more frequently, which would improve the realized volatility predictions in the sequential section. This is especially true for time periods like 2023 where the models would have quickly adapted to the lower volatility (the clustering of red dots in the 2023 section wouldn't be visible).

Additionally, the credible interval estimates for volatility would be more useful since there would be practically no measurement noise, which would make these models more competitively viable in a trading setting

7.2 Conclusions

In this project, the discrepancy between realized volatility in the regression and sequential highlights a fundamental challenge with testing these models and establishes why out-of-sample tests are crucial to determining a model's true strength.

In terms of the models themselves, the QLIKE loss result from rolling volatility highlights the need for more sophisticated volatility models.

Comparing the 2 sophisticated models, the Bayesian EGARCH model had a better overall performance than the Standard Stochastic Volatility model.

Its main advantage is its ability to deal with regime changes (particularly during more volatile periods). This result highlights the benefit of using models that incorporate well-known information about volatility. For instance, the leverage effect captured by the EGARCH model allowed it to better adapt to the sudden rise in volatility caused by the drastic fall in returns during COVID.

Furthermore, the EGARCH model was also well-calibrated for returns as shown by the p values in both the regression and sequential section, while the SV model didn't have the appropriate shape for its estimated return distributions.

Interestingly though, the stochastic volatility model was actually better at covering realized volatility during the regression task and was comparable to the EGARCH model during the sequential task despite the issues highlighted above. This occurred as stochastic volatility models tend to produce wider credible intervals since they factor in uncertainty through both the Bayesian framework and independent stochastic shocks in the volatility equations.

This analysis suggests that a stochastic model that incorporates more information about financial volatility alongside the independent stochastic shocks will probably perform better than the Bayesian EGARCH(1,1)-t model.

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